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Performance Assessment of Motion Tracking Methods in Ultrasound-based Shear Wave Elastography

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***Abstract*—Ultrasound elastography is a modality that is uniquely suited to augment conventional B-mode ultrasound for various clinical applications. Motion tracking plays a critically important role during image formation for ultrasound elastography. In this study, the accuracy of four motion tracking methods tailored for acoustic radiation force-based elastography (*e.g.* acoustic radiation force imaging, shear wave elastography) is compared. In these elastography methods, external mechanical excitation results in small tissue displacements (*i.e.* 5-10 micrometers). This paper compares four published motion tracking methods: a quadratic sub-sample estimation method, a coupled sub-sample estimation method, a 2-D spline-based estimator, and a 2-D autocorrelation-based motion estimator. Those four methods are evaluated using computer-simulated and tissue-mimicking phantom data. Based on our preliminary data, we find that the autocorrelation-based method is the preferred estimator without considering the lateral displacement. Overall, the spline-based estimator is superior to the other two competitors when both axial and lateral displacements are estimated. Since the spline-based estimation algorithm is considerably time-intensive, the coupled sub-sample estimation method becomes a practical alternative.**

***Index Terms*—ultrasound elastography, shear wave elastography, performance assessment, motion tracking, acoustic radiation force**

I. Introduction

Manual palpation is an effective method for detecting focal lesions and has been widely used for the breadth of modern medical history. Its effectiveness stems from the fact that changes in tissue stiffness directly reflect pathological evolution during disease processes. However, manual palpation is highly subjective and cannot provide quantitative information (*e.g.* size and location) of the pathology. The development of ultrasound elastography (UE) [1] has been motivated to overcome the above-mentioned limitations of manual palpation. It is worth noting that, before the invention of elastography, conventional medical modalities, such as X-ray, computed tomography (CT), magnetic resonance imaging (MRI), and ultrasound, cannot provide information directly related to tissue stiffness.

In recent years, under the common framework of UE, shear wave elastography (SWE) [2], [3] has gained rapid attention. SWE achieved successes from a range of clinical applications, including the staging of liver fibrosis and the detection of tumors in the prostate, thyroid, breast, and liver [4]–[6]. Moreover, it has also been used to evaluate chronic kidney disease [7], the musculoskeletal system [8]–[10], carotid atherosclerotic plaques [11], and the heart [12]. If we assume that soft tissues being imaged are linearly elastic as a first approximation, the measured shear wave speed (SWS) is nearly proportional to the square root of the local shear modulus. Since obtaining absolute physical measurements such as SWS offers unique advantages, major ultrasound vendors (*e.g.* GE, Philips, and Siemens) have capitalized on this opportunity by releasing their SWE packages.

In ultrasound SWE, acoustic pushes (*i.e.* focused ultrasound beams) is first used to excite soft tissues, thereby generating local shear waves. Due to concerns of bio-safety, only small tissue displacements (typically 5-10 micrometers) are generated by the above-mentioned acoustic pushes. Then, ultrasound data can be collected and tracked to measure the shear wave speed (SWS). More specifically, time-resolved local tissue motion is generally estimated through motion estimation (*e.g.* phase-based or correlation-based displacement estimation). Given time-resolved tissue motion, the shear wavefront can be tracked and SWS can be estimated using time-of-flight methods [13], [14]. In the time domain [13], shear wave arrival time can be linearly fitted with respect to (spatial) measurement positions along the wave propagation direction. Alternatively, in the frequency-domain, optimization can be done to find an SWS that best aligns all waveforms [14].

In this study, our primary objective is to assess how motion tracking accuracy influences SWS estimates. Estimating local tissue motion is critically important because errors among estimated tissue displacements compromise the accuracy of the SWS estimates. However, this aspect has often been overlooked largely because only small tissue motion is

induced by the above-mentioned acoustic pushes.
Many ultrasound-based motion tracking methods have been developed in the literature [15]. Consequently, exhaustive comparisons among them are difficult. We select four commonly used tracking methods: a 2-D quadratic sub-sample estimation algorithm [16], a 2-D spline-based estimator [17], a coupled subsample displacement estimation method [18], and a 2-D auto-correlation estimator [19]–[21]. Those four tracking methods are compared using computer-simulated SWE data and SWE data acquired from a tissue-mimicking phantom.

II. A Brief Summary of Four Selected Tracking Methods

In proceeding subsections, four selected tracking methods are briefly described for the sake of completeness. Given the sampling frequency (30-50 MHz) in modern clinical ultrasound scanners, small tissue displacements (5-10 micrometers) induced by acoustic pushes are often less than one acoustic sample along the axial direction. Thus, our description of all tracking methods focuses on sub-sample displacement estimation. Furthermore, in modern ultrasound scanners, radiofrequency (RF) ultrasound data are digitized. Given small displacements, correlation function between a pair of (discretely sampled) pre- and post-deformation signals can be easily calculated, *e.g.* through block-matching.

A. 2-D Quadratic Sub-sample Estimation Algorithm [16]

Given a pair of the pre- and post-deformation ultrasound signals, s_1 and s_2, $\rho(x,y)$ is a discrete normalized cross-correlation function as follow:

$$\begin{aligned}\rho(x,y) = &\frac{\sum\limits_{i=0}^{Q-1}\sum\limits_{j=0}^{R-1} s_1(i,j)\, s_2(i+x,j+y)}{\sqrt{\sum\limits_{i=0}^{Q-1}\sum\limits_{j=0}^{R-1} s_1^2(i,j)\sum\limits_{i=0}^{Q-1}\sum\limits_{j=0}^{R-1} s_2^2(i+x,j+y)}}\\ x =&0,1,\ldots,(M-Q)-1\\ y =&0,1,\ldots,(N-R)-1\end{aligned} \tag{1}$$

where $Q \times R$ and $M \times N$ ($M > Q$, $N > R$) are signal dimensions along the axial and lateral directions, respectively. In Eqn. (1), x and y are shifts during the calculation of $\rho(x,y)$. In this study, s_1 and s_2 are two-dimensional signals.
In this method, once the normalized cross-correlation function (see Eqn. (1)) is evaluated, a 2-D polynomial function is used to fit all correlation values over a 3×3 grid centered at the correlation peak. The polynomial function reads,

$$\rho(x,y) = a_1 + a_2 x + a_3 y + a_4 xy + a_5 x^2 + a_6 y^2 \tag{2}$$

In order to obtain the true correlation peak, the problem can be solved as a two-step process. First, 9 correlation values and their coordinates are used to obtain a_1 through a_6 in a least-square fashion. Then, the true peak locates at $\frac{\partial\rho(x,y)}{\partial x} = 0$ and $\frac{\partial\rho(x,y)}{\partial y} = 0$. Details can be found elsewhere [16].

B. 2-D Spline-Based Estimator [17]

Similarly, we denote s_1 and s_2 as a pair of the pre- and post-deformation ultrasound signals. In this method, s_1 is analytically represented by a series of piece-wise continuous polynomials. The bicubic polynomial function with 16 coefficients located at indices (i,j) is expressed as follows:

$$\begin{aligned}f_{i,j}(x,y) =& {}_0\kappa_{i,j} + {}_1\kappa_{i,j}x + {}_2\kappa_{i,j}y + {}_3\kappa_{i,j}x^2\\ &+ {}_4\kappa_{i,j}y^2 + {}_5\kappa_{i,j}xy + {}_6\kappa_{i,j}x^3\\ &+ {}_7\kappa_{i,j}y^3 + {}_8\kappa_{i,j}x^2y + {}_9\kappa_{i,j}xy^2\\ &+ {}_{10}\kappa_{i,j}x^3y + {}_{11}\kappa_{i,j}xy^3 + {}_{12}\kappa_{i,j}x^2y^2\\ &+ {}_{13}\kappa_{i,j}x^3y^2 + {}_{14}\kappa_{i,j}x^2y^3 + {}_{15}\kappa_{i,j}x^3y^3\end{aligned} \tag{3}$$

where $f(x,y)$ is a smooth and continuous representation of the original signal s_1.
In the process of estimating displacement, the last x^3y^3 term in the right hand side of Eqn. 3 is "eliminated" to enforce a smoothness constraint in the resultant displacement field. Formally, the highest-order term in Eqn. (3) is set to 0, *i.e.* ${}_{15}\kappa_{i,j} = 0$.
Mathematically, the 2-D spline representation with 15 parameters (*i.e.* $f(x,y)$) can be formulated as multiplications of a series of 1-D splines. Thus, the sum squared error (SSE) between $f(x,y)$ and $s_2(x,y)$ is minimized to estimate displacement, which is described as

$$\begin{aligned}E_{l,m}(x,y) = &\sum_{i=1}^{Q}\sum_{j=1}^{R}\left(f_{i+l,j+m}(x,y) - s_2[i,j]\right)^2\\ l =&0,1,\ldots,(M-Q)-1\\ m =&0,1,\ldots,(N-R)-1\end{aligned} \tag{4}$$

where $f_{i,j}(x,y)$ is the polynomial representation of s_1 at (i,j) as described in Eqn. (3). s_2 is the reference data set of dimensions $Q \times R$, and l and m are the integer sample offsets in axial and lateral directions, respectively. More details can be found the work by Viola *et al.* [17].

C. Coupled Subsample Displacement Estimation Method [18]

According to linear systems theory(*e.g.* [22]), a log-compressed cross-correlation function $\rho(x,y)$ between the pre- and post-deformed ultrasound signals can be written as

$$\begin{cases}\log(\rho[x,y]) \approx \frac{(x-dx)^2}{C_1} + \frac{(y-dy)^2}{C_2}\\ \frac{(x-dx)^2}{C_1} + \frac{(y-dy)^2}{C_2} = K\end{cases} \tag{5}$$

where C_1 and C_2 are related to the ultrasound system parameters but are two constants for two given pre- and post-deformed signals, and x and y denote a uniform grid around the correlation peak at the integer level. Similarly, the discrete correlation function $\rho(x,y)$ can be obtained using Eqn. (1). Derivations of Eqn. (5) can be found in the original publication [18].
As seen from Eqn. (5), an iso-contour of log-compressed correlation function is an ellipse, and the correlation peak is located at the center of this ellipse. Thus, Eqn. (5) provides an alternative to estimate the true peak of the correlation function.

$$\bar{\mu} = \frac{c}{4\pi f_c} \frac{\tan^{-1}\left\{\frac{\sum_{m=0}^{M-1}\sum_{n=0}^{N-2}\left[Q(m,n)\,I(m,n+1) - I(m,n)\,Q(m,n+1)\right]}{\sum_{m=0}^{M-1}\sum_{n=0}^{N-2}\left[I(m,n)\,I(m,n+1) + Q(m,n)\,Q(m,n+1)\right]}\right\}}{1 + \frac{1}{2\pi f_{dem}}\tan^{-1}\left\{\frac{\sum_{m=0}^{M-2}\sum_{n=0}^{N-1}\left[Q(m,n)\,I(m+1,n) - I(m,n)\,Q(m+1,n)\right]}{\sum_{m=0}^{M-2}\sum_{n=0}^{N-1}\left[I(m,n)\,I(m+1,n) + Q(m,n)\,Q(m+1,n)\right]}\right\}} \tag{7}$$

More specifically, estimating the true correlation peak invovles the following three steps. In the first step, an iso-contour at the correlation value K can be empirically determined as follows,

$$K = (\rho_{i-1,j-1} + \rho_{i-1,j} + \rho_{i-1,j+1} + \rho_{i,j-1} + \rho_{i,j+1} + \rho_{i+1,j-1} + \rho_{i+1,j} + \rho_{i+1,j+1})\,/8 \tag{6}$$

where $\rho_{i,j}$ is the correlation peak value located at the ith row and jth column of the correlation map. In practice, the selected K value is the averaged correlation value among the eight neighborhood of the peak value $\rho_{i,j}$ at integer level. In the second step, an iso-contour method (*e.g.* matching cube) can be used to determine coordinates of the iso-contour. In the third step, coordinates of the iso-contour are fitted into the ellipse using a least-squares approach to find the center of the fitted ellipse. As stated before, the center of the fitted ellipse (dx, dy) mathematically locates at the true peak of the correlation function, based on the linear system theory of ultrasound [22].

D. 2-D Autocorrelator [19]

The auto-correlation tracking method is proposed by Loupas $et\ al.$ for estimating axial displacements utilizing in-phase and quadrature (IQ) components of ultrasound RF data (hereafter referred to as IQ data) [20], [21]. It reads (see Eqn. (7)),

where M is the axial averaging range, m is the m^{th} axial sample being used, N is the ensemble length, n is the n^{th} IQ line being used, c is the speed of ultrasound, f_c is the center frequency, f_{dem} is the demodulation frequency, which is equal to f_c/f_s, f_s is the sampling RF rate, $\bar{\mu}$ is the average displacement over a given axial range, and I and Q are the in-phase and quadrature components of RF signal, which can be obtained by trigonometric quadrature demodulation in this paper. It's worth to note that at least two axial samples are required to obtain an estimate of the mean RF frequency.

III. Experimental Design

A. Computer-simulated SWE data

In the first dataset, 70 frames of computer-simulated ultrasound RF echo signals are used to evaluate the performance of the four above-said motion tracking methods. All calculations were preformed in MATLAB (MathWorks Inc., Natick, MA). Finite element models reported in a published study by Palmeri *et al.* are used to excite a uniform numerical phantom whose Young's modulus is approximately 28-30 kPa and dimensions of 100 mm×25 mm×10 mm [23]. The simulated acoustic push is focused at 30 mm, $F/N = 2$, and push duration is 334 μs. The transducer parameters used for Field II simulations are summarized in Table I.

TABLE I
THE CHANGED TRANSDUCER PARAMETERS FOR THE SIMULATION IN ELASTIC MEDIA

Parameter	Value
Center Frequency	7.56MHz
Sampling Frequency	100MHz
Attenuation	0.5dB/cm/MHz
Bandwidth	50%
Elevation Focus	50mm

The second dataset is obtained from a tissue-mimicking commercial Elasticity QA phantom (Model 49A, CISR Inc., VA, USA). Shear waves are generated by using acoustic pushing pulse. This process is done using a research ultrasound scanner (V1, Verasonics Inc., WA, USA) and more detailed data acquisition parameters can be found in our previous publications [14], [24]. The same scanner is used to acquire ultrasound data containing shear wave motion for the above-mentioned motion tracking. Correlation calculations in the coupled, quadratic and auto-correlation methods are done using sum-table method [25] to improve computational efficiency if possible.

B. Data Analysis

In addition to qualitative assessments, mean and variances of tracking errors are computed. The tracking error is defined as difference between the estimated and ground truth displacements as follows,

$$error = (d_{es} - d_{gt}) \tag{8}$$

where d_{es} and d_{gt} are the estimated and ground truth displacements, respectively. The mean error is an indicator of tracking bias while the error variances become a metric of tracking precision.

All data analyses including computational efficiency are done using a computer workstation equipped with a single CPU (i7@2.6GHz, Intel Inc., Santa Clara, CA, USA).

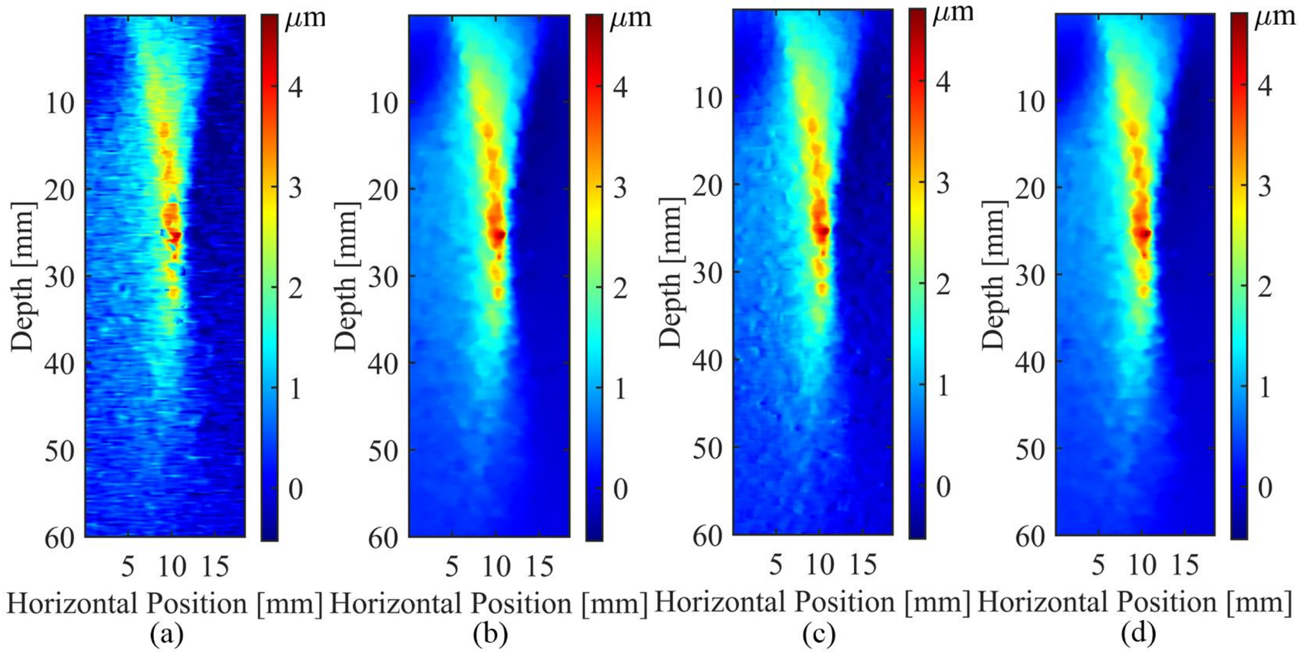

Fig. 1. Axial displacement images between the frames 0 and 40 of the first dataset using four motion tracking methods. These images are obtained by (a) 2-D quadratic sub-sample estimation algorithm, (b) 2-D spline-based estimator, (c) coupled sub-sample displacement estimation method and (d) 2-D autocorrelator, respectively.

IV. RESULTS

Table II depicts the mean and variance of tracking errors for the computer-simulated phantom. The quadratic sub-sample method has a slightly higher mean error value as compared to the other three methods, as shown in Table II. Typically, displacements induced by acoustic pushes are approximately 10-15 μm. Thus, the estimation biases represented by mean errors are low ($< 0.2\%$; Table II) and all four methods can be practically considered as unbiased estimators in terms of tracking small displacements. All four methods also show comparable error variances.

TABLE II
THE MEAN OF THE ESTIMATED ERRORS AND VARIANCES

	Error (μm)	Variance (μm^2)
2-D Quadratic	-0.0213	0.5911
2-D Spline	0.0099	0.6045
Coupled	0.0062	0.6250
Autorcorrelator	0.0045	0.5856

Overall, the autocorrelator has the lowest mean error and error variance. That explains the broad adoption of this autocorrelator method in the acoustic radiation-based ultrasound elastography.

Visually, the autocorrelator (Fig. 1(d)) and spline-based method (Fig. 1(b)) provide smoother displacement estimates. One representative example of the estimated axial displacement image (between the frames 0 and 40 in simulated data) is shown in Fig. 1.

After all displacements has been obtained, the wavefronts (*i.e.* peak displacements along the lateral direction over time) are used to calculate shear wave speed (SWS) using a published time-to-peak method [13]. Then, the SWS values are converted to Young's modulus. Fig. 2 compares the estimated moduli from the tissue-mimicking phantom at different depths, which are derived from the estimated displacements of the above-mentioned four motion tracking methods. Again, Young's modulus values obtained using displacements estimated by the above-said four tracking methods are comparable, though the autocorrelator provides the most stable results.

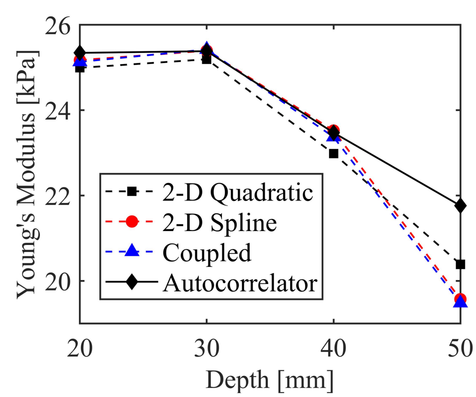

Fig. 2. A plot of estimated Young's Modulus values at different depths using a lateral time-to-peak (TTP) algorithm based on RANSAC filtering [13]

Fig. 3 shows the lateral and axial displacement images estimated by these four motion tracking methods using the tissue-mimicking phantom data. It can be seen that the estimated axial displacement results of the autocorrelator (Fig. 3(g)) and spline-based tracking method (Fig. 3(d)) are smoother as compared to the other two methods. However, the autocorrelator cannot estimate lateral displacements. Thus, the lateral displacements obtained using the spline-based estimator (Fig. 3(c)) are smoother than those of the two methods. Qualitatively, the axial and lateral displacement images estimated by the quadratic subsample estimation algorithm (Fig. 3(a) and Fig. 3(b)) are relatively noisy, thereby suggesting that its displacement estimates are lower in quality than the other three methods.

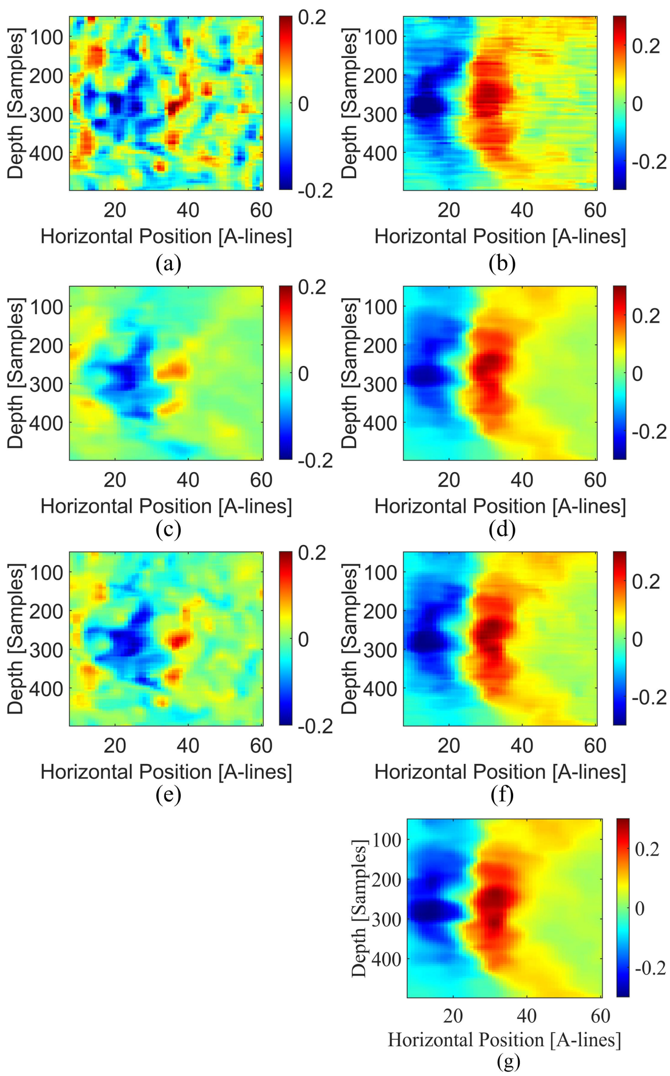

Fig. 3. The lateral (left column) and axial (right column) displacement images estimated by four motion tracking methods using ultrasound data acquired from the tissue-mimicking phantom. The images obtained by the 2-D quadratic subsample estimation algorithm, 2-D spline-based estimator, coupled subsample motion tracking method and 2-D autocorrelator, are shown in the rows 1-4, respectively.

The estimated frame rates for the four tracking methods can be seen in Table III.

TABLE III
A SUMMARY OF PROJECTED FRAME RATES FOR FOUR DIFFERENT TRACKING METHODS. THE FRAME RATES ARE ESTIMATED BASED ON TIME NEEDED TO PRODUCE DISPLACEMENTS ON A 100×100 GRID.

Tracking Method	Frame Rate (f/s)
Quadratic	1.687
Spline	0.003
Coupled	0.2625
Auto-correlation	4.030

V. DISCUSSION AND SUMMARY

A comparison of four motion tracking methods is performed using computer-simulated ultrasound data and data acquired from a tissue-mimicking phantom. Based on our preliminary results, the 2-D autocorrelator has advantages for estimating axial displacement for SWE applications if there is no need to compensate for lateral displacements. According to the aggregated results presented above, the 2-D quadratic subsample method demonstrates the lowest performance. However, in many clinical applications, tracking lateral motion is necessary to compensate for physiological motion in organs affected due to cardiac and respiratory motion. Consequently, the spline-based and coupled tracking method can be two viable options. Although the spline-based method yields smooth displacements, it is computationally expensive. Consequently, the coupled tracking method may offer a balance in terms of computational efficiency and performance. Also, GPU-acceleration of motion tracking methods [26], [27] should be carefully investigated and remains as our ongoing research plan.

It is also important to note that our investigations are limited to motion tracking of shear wave elastography in simulated and tissue-mimicking phantom data. Thus, our conclusions are bounded by data investigated. In the presence of *in vivo* physiological motion during the shear wave elastography data acquisition, motion tracking algorithms must be able to track large motion. Similarly, accurately tracking of large tissue deformation is also required for the estimation of nonlinear mechanical properties of soft tissues [10], [24], [28], [29]. A survey of motion tracking for elastography applications can be found elsewhere [30].

In our recent work, we found that neural network-based tracking [31] reliably tracked *in vivo* gross tissue motion given good training data by ultrasound simulations [32], [33]. Our future plan is to combine neural network-based speckle tracking [31], [34] with sophisticated sub-sample tracking algorithms to improve *in vivo* shear wave elastography.

REFERENCES

[1] J. Ophir, I. Cespedes, H. Ponnekanti, Y. Yazdi, and X. Li, "Elastography: A Quantitative Method for Imaging the Elasticity of Biological Tissues," *Ultrasonic Imaging*, vol. 13, no. 2, pp. 111–134, 1991.

[2] A. Sarvazyan, T. J Hall, M. W Urban, M. Fatemi, S. R Aglyamov, and B. S Garra, "An Overview of Elastography-an Emerging Branch of Medical Imaging," *Current Medical Imaging Reviews*, vol. 7, no. 4, pp. 255–282, 2011.

[3] T. Shiina, K. R. Nightingale, M. L. Palmeri, T. J. Hall, J. C. Bamber, R. G. Barr, L. Castera, B. I. Choi, Y.-H. Chou, D. Cosgrove *et al.*, "WFUMB Guidelines and Recommendations for Clinical Use of Ultrasound Elastography: Part 1: Basic Principles and Terminology," *Ultrasound in Medicine & Biology*, vol. 41, no. 5, pp. 1126–1147, 2015.

[4] R. G. Barr, R. Memo, and C. R. Schaub, "Shear Wave Ultrasound Elastography of the Prostate: Initial Results," *Ultrasound Quarterly*, vol. 28, no. 1, pp. 13–20, 2012.

[5] J. Bamber, D. Cosgrove, C. Dietrich, J. Fromageau, J. Bojunga, F. Calliada, V. Cantisani, J.-M. Correas, M. D'onofrio, E. Drakonaki *et al.*, "EFSUMB Guidelines and Recommendations on the Clinical Use of Ultrasound Elastography. Part 1: Basic Principles and Technology," *Ultraschall in der Medizin-European Journal of Ultrasound*, vol. 34, no. 02, pp. 169–184, 2013.

[6] N. Frulio, H. Laumonier, T. Carteret, C. Laurent, F. Maire, C. Balabaud, P. Bioulac-Sage, and H. Trillaud, "Evaluation of Liver Tumors Using Acoustic Radiation Force Impulse Elastography and Correlation with Histologic Data," *Journal of Ultrasound in Medicine*, vol. 32, no. 1, pp. 121–130, 2013.

[7] Q. Hu, X.-Y. Wang, H.-G. He, H.-M. Wei, L.-K. Kang, and G.-C. Qin, "Acoustic Radiation Force Impulse Imaging for Non-invasive Assessment of Renal Histopathology in Chronic Kidney Disease," *PloS One*, vol. 9, no. 12, p. e115051, 2014.

[8] A. Nordez and F. Hug, "Muscle Shear Elastic Modulus Measured Using Supersonic Shear Imaging Is Highly Related to Muscle Activity Level," *Journal of applied physiology*, vol. 108, no. 5, pp. 1389–1394, 2010.

[9] S. F. Eby, B. A. Cloud, J. E. Brandenburg, H. Giambini, P. Song, S. Chen, N. K. LeBrasseur, and K.-N. An, "Shear Wave Elastography of Passive Skeletal Muscle Stiffness: Influences of Sex and Age Throughout Adulthood," *Clinical Biomechanics*, vol. 30, no. 1, pp. 22–27, 2015.

[10] R. J. DeWall, J. Jiang, J. J. Wilson, and K. S. Lee, "Visualizing Tendon Elasticity in an ex Vivo Partial Tear Model," *Ultrasound in Medicine & Biology*, vol. 40, no. 1, pp. 158 – 167, 2014.

[11] K. V. Ramnarine, J. W. Garrard, B. Kanber, S. Nduwayo, T. C. Hartshorne, and T. G. Robinson, "Shear Wave Elastography Imaging of Carotid Plaques: Feasible, Reproducible and of Clinical Potential," *Cardiovascular Ultrasound*, vol. 12, no. 1, p. 49, 2014.

[12] P. Song, H. Zhao, M. W. Urban, A. Manduca, S. V. Pislaru, R. R. Kinnick, C. Pislaru, J. F. Greenleaf, and S. Chen, "Improved Shear Wave Motion Detection Using Pulse-inversion Harmonic Imaging with a Phased Array Transducer," *IEEE Transactions on Medical Imaging*, vol. 32, no. 12, pp. 2299–2310, 2013.

[13] M. H. Wang, M. L. Palmeri, V. M. Rotemberg, N. C. Rouze, and K. R. Nightingale, "Improving the Robustness of Time-of-flight Based Shear Wave Speed Reconstruction Methods Using RANSAC in Human Liver in Vivo," *Ultrasound in Medicine & Biology*, vol. 36, no. 5, pp. 802–813, 2010.

[14] D. Rosen and J. Jiang, "Fourier-Domain Shift Matching: A Robust Time-of-Flight Approach for Shear Wave Speed Estimation," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 65, no. 5, pp. 729–740, May 2018.

[15] J. Jiang and B. Peng, *Ultrasonic Methods for Assessment of Tissue Motion in Elastography*. John Wiley & Sons, Ltd, 2018, ch. 4, pp. 35–70.

[16] R. Z. Azar, O. Goksel, and S. E. Salcudean, "Sub-sample Displacement Estimation from Digitized Ultrasound RF Signals Using Multi-dimensional Polynomial Fitting of the Cross-correlation Function," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 57, no. 11, pp. 2403–2420, 2010.

[17] F. Viola, R. L. Coe, K. Owen, D. A. Guenther, and W. F. Walker, "Multi-dimensional Spline-based Estimator (MUSE) for Motion Estimation: Algorithm Development and Initial Results," *Annals of Biomedical Engineering*, vol. 36, no. 12, p. 1942, 2008.

[18] J. Jiang and T. J. Hall, "A Coupled Subsample Displacement Estimation Method for Ultrasound-based Strain Elastography," *Physics in Medicine & Biology*, vol. 60, no. 21, p. 8347, 2015.

[19] T. Loupas, J. Powers, and R. W. Gill, "An Axial Velocity Estimator for Ultrasound Blood Flow Imaging, Sased on a Full Evaluation of the Doppler Equation by Means of a Two-dimensional Autocorrelation Approach," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 42, no. 4, pp. 672–688, 1995.

[20] G. F. Pinton, J. J. Dahl, and G. E. Trahey, "Rapid Tracking of Small Displacements with Ultrasound," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency control*, vol. 53, no. 6, pp. 1103–1117, 2006.

[21] S. Rosenzweig, M. Palmeri, and K. Nightingale, "GPU-based Real-Time Small Displacement Estimation with Ultrasound," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency control*, vol. 58, no. 2, pp. 399–405, 2011.

[22] R. J. Zemp, C. K. Abbey, and M. F. Insana, "Linear System Models for Ultrasonic Imaging: Application to Signal Statistics," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 50, no. 6, pp. 642–654, June 2003.

[23] M. L. Palmeri, B. Qiang, S. Chen, and M. W. Urban, "Guidelines for Finite-element Modeling of Acoustic Radiation Force-induced Shear Wave Propagation in Tissue-mimicking Media," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency control*, vol. 64, no. 1, pp. 78–92, 2016.

[24] D. P. Rosen and J. Jiang, "A Comparison of Hyperelastic Constitutive Models Applicable to Shear Wave Elastography (SWE) Data in Tissue-mimicking Materials," *Physics in Medicine & Biology*, vol. 64, no. 5, p. 055014, mar 2019.

[25] B. Peng, S. Luo, Z. Xu, and J. Jingfeng Jiang, "Accelerating 3-D GPU-based Motion Tracking for Ultrasound Strain Elastography Using Sum-Tables: Analysis and Initial Results," *Applied Science*, vol. 9, no. 10, p. 1991, mar 2019.

[26] B. Peng, Y. Wang, T. J. Hall, and J. Jiang, "A GPU-Accelerated 3-D Coupled Subsample Estimation Algorithm for Volumetric Breast Strain Elastography," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 64, no. 4, pp. 694–705, 2017.

[27] B. Peng, S. Luo, Z. Xu, and J. Jiang, "Accelerating 3-D GPU-based Motion Tracking for Ultrasound Strain Elastography Using Sum-Tables: Analysis and Initial Results," *Applied Sciences*, vol. 9, no. 10, 2019.

[28] T. J. Hall, A. A. Oberait, P. E. Barbone, A. M. Sommer, N. H. Gokhale, S. Goenezent, and J. Jiang, "Elastic nonlinearity imaging," in *2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2009, pp. 1967–1970.

[29] D. Rosen and J. Jiang, "Analyzing acoustoelastic effect of shear wave elastography data for perfused and hydrated soft tissues using a macromolecular network inspired model," *Journal of Biomechanics*, vol. 97, p. 109370, 2019.

[30] J. Jiang and B. Peng, *Ultrasonic Methods for Assessment of Tissue Motion in Elastography*. John Wiley & Sons, Ltd, 2018, ch. 4, pp. 35–70.

[31] B. Peng, Y. Xian, Q. Zhang, and J. Jiang, "Neural-network-based motion tracking for breast ultrasound strain elastography: An initial assessment of performance and feasibility," *Ultrasonic Imaging*, vol. 42, no. 2, pp. 74–91, 2020.

[32] Y. Wang, E. Helminen, and J. Jiang, "Building a virtual simulation platform for quasistatic breast ultrasound elastography using open source software: A preliminary investigation," *Medical Physics*, vol. 42, no. 9, pp. 5453–5466, 2015.

[33] Y. Wang, B. Peng, and J. Jiang, "Building an open-source simulation platform of acoustic radiation force-based breast elastography," *Physics in Medicine and Biology*, vol. 62, no. 5, pp. 1949–1968, feb 2017.

[34] L. He, B. Peng, T. Yang, and J. Jiang, "An application of super-resolution generative adversary networks for quasi-static ultrasound strain elastography: A feasibility study," *IEEE Access*, vol. 8, pp. 65 769–65 779, 2020.